

Stock Market App

Team Name: Best Team

Purpose:

The purpose of our Stock Market App is to help users better understand the stock market, track investments, and make informed financial decisions in an accessible way. Many beginner investors are interested in stocks but feel overwhelmed by confusing market data, changing prices, and financial language that is hard to fully understand. Our application aims to solve that problem by organizing stock information into a clearer, more user-friendly dashboard that highlights important trends such as the greatest growing stocks, recommended stocks, and stocks that have recently dropped.

This project is important because more people are becoming interested in investing, but many don't have the tools or experience needed to confidently invest in the market. Existing financial platforms can feel too advanced for casual users or students who are still learning. Our app combines useful stock data with personalized features such as favorites, performance reports, a simulation account, and an AI chatbot to create a more supportive experience. The chatbot, powered by Gemini, will give users a way to ask questions about dips, gains, or market activity in plain language. Overall, the goal of our app is to make stock tracking, learning, and practice investing more approachable, educational, and engaging for everyday users.

Users:

The primary users of our application are beginner investors, college students, and casual market watchers who want an easier way to follow stocks and learn about investing. These users may have limited financial knowledge, little experience with trading platforms, or simply want a less **cut throat** environment to explore the stock market. **Some users may already have favorite stocks they want to monitor or put "fake"** money in, while others may want a place to practice investing without using real money. Our app is designed to meet both of these needs by offering real market updates alongside a simulation account for learning and experimentation.

This application will benefit users

Features:

Our Stock Market App includes several core features that work together to support both market tracking and investment practice. The home page will display categories such as greatest growing stocks, recommended stocks, and stocks that have recently dropped. This gives users a quick overview of what is happening in the market without needing to search. The Favorited Stocks page will allow users to save stocks they care about and view updates focused specifically on those selections. The Reports page will display a chart that contains the name of the stock, 24 change and a graphs based on the user's favorite stocks. Within the myStalks page there is a simulation activity, including profit and loss, changes in value, and indicators of whether the user's investing choices appear safe or risky.

Another major feature is the chatbot, which appears as a button in the corner of the screen and opens a chat window when clicked. This chatbot will use the Gemini API to answer questions about stock behavior.

Data:

The app handles data through standard CRUD operations split between local storage and the Finnhub API.

Users create data by adding stocks to their portfolio. Each entry is saved to local storage as an object containing the ticker symbol, number of shares, purchase price, and an object of transaction information. For example, adding Apple might look like: { ticker: "AAPL", shares: 10, transactions: [{ type: "buy", shares: 10, price: 189.50, date: "2025-03-30" }] }. The transaction object will contain the transaction type, quantity, price, and date in order to log multiple orders and sales.

Users read data two ways. Portfolio entries are pulled from local storage to populate the Favorites and Reports pages, while live market data like current price, percent change, and company details are read from the Finnhub API using each saved ticker and calling the proper endpoint.

Users update data by buying or selling shares of an existing portfolio entry. Each transaction appends a new entry to that stock's transaction log and updates the total share count accordingly. For example, a follow-up purchase of 5 more shares would add { type: "buy", shares: 5, price: 192.00, date: "2025-04-02" } to the log, bringing the total to 15 shares. This preserves full purchase history for use in the Reports page.

Users delete data by removing a stock from their portfolio, which clears that entry from local storage and removes it from all related pages.

